Document Identifier: **PMC8189437**

SciScore Report

Below you will find your SciScore report containing three tables. Your score is calculated based on adherence to scientific rigor criteria (Table 1) and identification of key biological resources (Table 2). Table 3 contains statistical tests and oligonucleotides but is not scored. If SciScore makes any mistakes, please contact us to help us learn and improve.

Table 1: Rigor Adherence Table

Ethics
not detected.
Inclusion and Exclusion Criteria
not required.
Attrition
not required.
Sex as a biological variable
not detected.
Subject Demographics
Age: not detected.
Weight: not detected.
Randomization
The missing values were subsequently replaced by random numbers that were drawn from a normal distribution (width = 0.3 and down shift = 1.8).
Blinding
not detected.
Power Analysis
not detected.
Replication
All the assays were performed in at least duplicates and some of them were quantified using image analysis software ImageQuant (GE healthcare; version 8.2).
Number: All the assays were performed in at least duplicates and some of them were quantified using image analysis software ImageQuant (GE healthcare; version 8.2).

Cell Line Authentication	
Contamination: Cell densities were kept between 0.1-1 x 106 cells/mL, and cultures were regularly checked for the absence of mycoplasma contamination.	
Data Information	
Availability: Materials availability All unique/stable reagents generated in this study are available from the lead contact with a completed Materials Transfer Agreement.	
Availability: All proteomics data have been deposited on ProteomeXchange with the dataset identifier PRIDE: PXD024462.	
Identifiers: All proteomics data have been deposited on ProteomeXchange with the dataset identifier PRIDE: PXD024462.	PXD024462
Identifiers: All the unprocessed image data have been deposited to Mendeley Data : http://dx.doi.org/10.17632/rfpsg6939c.1 Method details Yeast strain construction and growth conditions The yeast strains used in this study are specified in the Key Resources Table.	http://dx.doi.org/10.17632/rfpsg6939c.1

Table 2: Key Resources Table

Your Sentences	REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies			
STARMethods Key resources table REAGENT or RESOURCE SOURCE IDENTIFIER Antibodies Monoclonal ANTI-FLAG M2 antibody Sigma Aldrich Cat# F1804; RRID: AB_262044 Anti-HA antibody produced in rabbit Sigma Aldrich Cat# H6908; RRID: AB_260070 Goat anti-rabbit IgG Dylight488 conjugated Invitrogen Cat# 35552; RRID: AB_844398 Goat anti-mouse IgG Dylight633 conjugated Invitrogen Cat# 35512; RRID: AB_1307538 Anti-rabbit peroxidase antibody produced in goat Sigma Aldrich Cat# A9169; RRID: AB_258434 Anti-mouse IgG Peroxidase antibody produced in goat Sigma Aldrich Cat# A4416; RRID: AB_258167 Anti-His antibody produced in mouse Cell Signaling Technology Cat# 9991; RRID: AB_2797714 Goat polyclonal anti-RMND5A antibody Santa Cruz Cat# sc-161202, RRID: AB_2181510 Sheep polyclonal anti-MAEA antibody R&D Systems Cat# AF7288-SP, RRID: AB_10971438 Rabbit polyclonal anti-RANBP9 antibody Abnova Cat# PAB16671; RRID: AB_10677213 Rabbit polyclonal anti-TWA1 antibody Novus Cat# NBP1-32596; RRID: AB_2274921 Mouse monoclonal anti-ARMC8 antibody Santa Cruz Cat# sc-365307; RRID: AB_10850172 Mouse monoclonal anti-MKLN1 antibody Santa Cruz Cat# sc-398956; RRID: AB_2737249 Rabbit polyclonal anti-WDR26 antibody Bethyl Laboratories Cat# A302-245A; RRID: AB_1730876 Rabbit polyclonal anti-YPEL5 antibody Thermo Fisher Cat# PA5-26957; RRID: AB_2544457 Sheep polyclonal anti-hGid4 This study N/A HaloLink Resin Promega Cat# G1912 ANTI-FLAG M2 affinity gel Sigma Aldrich Cat# A2220 His-Select Nickel affinity gel Sigma Aldrich Cat# P6611 Glutathione	ANTI-FLAG		
	Monoclonal ANTI-FLAG M2	Sigma-Aldrich	(Sigma-Aldrich Cat# F1804, RRID:AB_262044)(link)
	Anti-HA in rabbit	Sigma-Aldrich	(Sigma-Aldrich Cat# H6908, RRID:AB_260070)(link)
	Goat IgG	Thermo Fisher Scientific	(Thermo Fisher Scientific Cat# 35552, RRID:AB_844398)(link)
	Goat anti-mouse IgG Dylight633	Thermo Fisher Scientific	(Thermo Fisher Scientific Cat# 35512, RRID:AB_1307538)(link)
	Anti-rabbit		
	Anti-rabbit peroxidase in goat	Sigma-Aldrich	(Sigma-Aldrich Cat# A9169, RRID:AB_258434)(link)
	Anti-mouse IgG Peroxidase in	Sigma-Aldrich	(Sigma-Aldrich Cat# A4416, RRID:AB_258167)(link)
	Anti-His in mouse	Cell Signaling Technology	(Cell Signaling Technology Cat# 9991, RRID:AB_2797714)(link)
	anti-RMND5A		
	Goat polyclonal anti-RMND5A	Santa Cruz Biotechnology	(Santa Cruz Biotechnology Cat# sc-161202, RRID:AB_2181510)(link)
	anti-MAEA		
	anti-MAEA	R and D Systems	(R and D Systems Cat# AF7288, RRID:AB_10971438)(link)
	polyclonal	Abnova	(Abnova Cat# PAB16671, RRID:AB_10677213)(link)
	anti-TWA1		
			(Novus Cat# NBP1-32596, RRID:AB_2274921)(link)
	anti-ARMC8	Santa Cruz Biotechnology	(Santa Cruz Biotechnology Cat# sc-365307, RRID:AB_10850172)(link)
	Mouse	Santa Cruz Biotechnology	(Santa Cruz Biotechnology Cat# sc-398956, RRID:AB_2737249)(link)
			(Thermo Fisher Scientific Cat# A302-245A, RRID:AB_1730876)(link)
	anti-YPEL5		

Sepharose 4B GE Healthcare Cat# 17075605 StrepTactin Sepharose High Performance resin cytiva Cat# 28935599 His-Halo UBAUBQLN1 This study N/A Critical commercial assays EnzChek Phosphate Assay Kit ThermoFisher Scientific Cat# E6646 Bacterial and virus strains E. coli BL21 RIL (DE3) MPIB N/A E. coli DH5a MPIB N/A Chemicals, peptides, and recombinant proteins complete EDTA free Roche Cat# 05056489001 Aprotinin from bovine lung Sigma A1153-10MG Leupeptin Sigma L2884-250MG Benzamidine Sigma B6506-25G GGGGGFYVK-FAM MPIB N/A PTLVNGWPR MPIB N/A PTLVNGPRRDSTEGFTGRGWSGRGWSKGGK-FAM MPIB N/A PGLWRSPRRDSTEGFTGRGWSGRGWSKGGK-FAM MPIB N/A 3xFLAG peptide MPIB N/A Deposited data Apo Chelator-GIDSR4 This study EMDB: EMD-12541 Chelator-GIDSR4 + Fbp1 This study EMDB: EMD-12557 GIDSR4 This study EMDB: EMD-12548 SRS module This study EMDB: EMD-12559; PDB: 7NS3 Cat module This study EMDB: EMD-12560; PDB: 7NS4 SA module This study EMDB: EMD-12563; PDB: 7NSB Endogenous GIDAnt This study EMDB: EMD-12538 Endogenous Chelator-GIDAnt This study EMDB: EMD-12540 CTLHSR4 This study EMDB: EMD-12537 CTLHSR4 SRS module This study EMDB: EMD-12564; PDB: 7NSC CTLH-WDR26 SA and SRS modules This study EMDB: EMD-12545 CTLH-WDR26 supramolecular assembly This study EMDB: EMD-12542 CTLH-MKLN1 SA and SRS modules This study EMDB: EMD-12547 Fbp1 (crystal structure) This study PDB: 7NS5 Proteomics data This study PXD024462 Raw image data This study http://dx.doi.org/10.17632/rfpsg6939c.1 Experimental models: Cell lines Sf9 Insect cells Thermo Fisher Cat# 11496015 High Five Insect cells Thermo Fisher	polyclonal	Thermo Fisher Scientific	(Thermo Fisher Scientific Cat# PA5-26957, RRID:AB_2544457)(link)
	anti-hGid4		
	Fbp1		
	RANBP9		
	ARMC8		

Cat# B85502 K562 human cells ATCC ATCC#CCL-243; RRID: CVCL_00004 Experimental models: Organisms/strains Saccharomyces cerevisiae: Strain S288C: BY4741; MATa his3D1leu2D0 met15D0 ura3D0 Euroscarf Cat# Y00000 CRLY12; BY4741, Gid4::KANMX This study N/A CRLY14; BY4741, Gid7::KANMX This study N/A CRLY45; BY4741, Gid8::Gid8-3xFLAG-KANMX Qiao et. al., 2020 N/A CRLY131; BY4741, Gid2::3xFLAG-Gid2 (K365A) Qiao et. al., 2020 N/A CRLY241; BY4741, Gid7::KANMX, Gid8::Gid8-3xFLAG-HPHNT1 This study N/A CRLY267; BY4741, Gid7::Gid7-3xHA-HPHNT1, Gid5::Gid5-3xFLAG-KANMX This study N/A CRLY498; BY4741, Fbp1::Fbp1-3xFLAG-HPHNT1, Pdr5::NATNT2 This study N/A CRLY504; BY4741, Fbp1::Fbp1-3xFLAG-HPHNT1, Pdr5::NATNT2, Gid7:KANMX This study N/A Recombinant DNA pCSJ95 Chen et al., 2017 N/A pCSJ125 Chen et al., 2017 N/A VBP6; pRS313-pGPD-Pck1-3xFLAG-CYC-pGPD-DHFR-HA-CYC This study N/A DSJC1; pRS313-pGPD-Fbp1-K32R/K35R-3xFLAG-CYC-pGPD-DHFR-HA-CYC This study N/A DSJC2; pRS313-pGPD-Fbp1-K280R/K281R-3xFLAG-CYC-pGPD-DHFR-HA-CYC This study N/A DSJC3; pRS313-pGPD-Fbp1-K32R/K35R/K280R/K281R-3xFLAG-CYC-pGPD-DHFR-HA-CYC This study N/A pRS415-pTEF-CYC This study N/A pRS415-pTEF-GFP-ScGid4-CYC This study N/A pLIB Gid1 This study N/A pLIB Gid2 This study N/A pLIB Gid4 This study N/A pLIB Gid5 This study N/A pLIB Gid7 This study N/A pLIB Gid8-TEV-2xStrep This study N/A pLIB Gid8 This study N/A pLIB Gid9 This study N/A pLIB RANBP9 This study N/A pLIB RMND5A This study N/A pLIB hGid4 This study N/A pLIB ARMC8 This study N/A pLIB 2xStrep-3C-ARMC8 This study N/A pLIB MAEA This study N/A pLIB WDR26 This study N/A				

pLIB GST-TEV-WDR26 This study N/A pLIB MKLN1 This study N/A pLIB YPEL5 This study N/A pLIB MAEA Y394A This study N/A pLIB RMND5A R340A This study N/A pLIB RMND5A I338A/L339A This study N/A pLIB GST-TEV-Uba1 This study N/A pBIG2 Gid1:Gid8-TEV-2xS:Gid5:Gid4:Gid2:Gid9 This study N/A pBIG2 Gid1:Gid8-TEV-2xS:Gid5:Gid2:Gid9 This study N/A pBIG2 Gid1:Gid8-TEV-2xS:Gid5:Gid2:Gid9:Gid7 This study N/A pBIG2 Gid1:Gid8-TEV-2xS:Gid5:Gid4:Gid2:Gid9:Gid7 This study N/A pBIG1 RANBP9:TWA1-TEV-2xS:ARMC8 This study N/A pBIG1 RANBP9:TWA1:2xS-3C-ARMC8 This study N/A pBIG2 RANBP9:TWA1-TEV-2xS:ARMC8:RMND5A:MAEA This study N/A pBIG2 RANBP9:TWA1-TEV-2xS:ARMC8:RMND5A This study N/A pBIG2 RANBP9:TWA1-TEV-2xS:ARMC8:MAEA This study N/A pGEX GST-TEV-Gid4 (D1-116) This study N/A pGEX GST-TEV-Gid7 This study N/A pGEX GST-TEV-Gid7 (D1-285) This study N/A pGEX GST-TEV-hGid4 (D1-99) This study N/A pGEX GST-TEV-hGid4 (R189A) This study N/A pGEX GST-TEV-hGid4 (Y154A) This study N/A pGEX GST-TEV-hGid4 (F174A) This study N/A pGEX GST-TEV-hGid4 (F239A) This study N/A pGEX GST-TEV-hGid4 (C156D) This study N/A pGEX GST-TEV-hGid4 (E298A) This study N/A pGEX GST-TEV-hGid4 (H147D) This study N/A pGEX GST-TEV-hGid4 (Y158A/F174A) This study N/A pGEX GST-TEV-hGid4 (Y158A/F239A) This study N/A pGEX GST-TEV-hGid4 (F174A/F239A) This study N/A pGEX GST-TEV-hGid4 (Y297A/F229A) This study N/A pGEX GST-TEV-hGid4 (D297-300) This study N/A pRSF Ubc8-6xHis This study N/A pRSF Fbp1-6xHis This study N/A pRSF Fbp1 (K32R/K35R)-6xHis This study N/A pRSF Fbp1 (K280R/K281R)-6xHis This study N/A pRSF Fbp1 (K32R/K35R/K280R/K281R)-6xHis This				

study N/A pRSF Fbp1-GGGGS-sortag-6xHis This study N/A pRSF Mdh2-GGGGS-sortag-6xHis This study N/A pRSF Pck1-GGGGS-sortag-6xHis This study N/A pQlink Fbp1-TEV-V5-2xS This study N/A pRSF Ube2H-6xHis This study N/A pGEX GST-3C-Ub This study N/A pGEX GST-3C-Ub K0 (all K > R) This study N/A pGEX GST-3C-Ub K6 (all K > R; R6K) This study N/A pGEX GST-3C-Ub K11 (all K > R; R11K) This study N/A pGEX GST-3C-Ub K27 (all K > R; R27K) This study N/A pGEX GST-3C-Ub K29 (all K > R; R29K) This study N/A pGEX GST-3C-Ub K33 (all K > R; R33K) This study N/A pGEX GST-3C-Ub K48 (all K > R; R48K) This study N/A pGEX GST-3C-Ub K63 (all K > R; R63K) This study N/A pET3b Ub This study N/A pET29 sortase A Chen et al., 2011b N/A				
Software and algorithms FOCUS Biyani et al., 2017 https://focus.c-cina.unibas.ch/documentation.php SerialEM Mastronarde, 2003 https://bio3d.colorado.edu/SerialEM/ MOTIONCOR2 Zheng et al., 2017 https://emcore.ucsf.edu/ucsf-software Gctf Zhang, 2016 https://www2.mrc-lmb.cam.ac.uk/download/gctf/ Gautomatch Kai Zhang https://www2.mrc-lmb.cam.ac.uk/download/gautomatch-053/ Relion3.0/3.1 Fernandez-Leiro and Scheres, 2017; Scheres, 2012; Zivanov et al., 2018 https://www3.mrc-lmb.cam.ac.uk/relion/index.php/Main_Page Phyre2 Kelley et al., 2015 http://www.sbg.bio.ic.ac.uk/~phyre2/html/page.cgi?id=index SWISS-MODEL Waterhouse et al., 2018 https://swissmodel.expasy.org UCSF Chimera Pettersen et al., 2004 https://www.cgl.ucsf.edu/chimera/ UCSF ChimeraX Goddard et al., 2018 https://www.rbvi.ucsf.edu/chimerax/ PyMOL v2.1 Schrodinger https://pymol.org/2/ CCP-EM Burnley et al., 2017 https://www.ccpem.ac.uk/download.php Buccaneer Cowtan, 2006 http://www.ysbl.york.ac.uk/~cowtan/buccaneer/buccaneer.html Coot Emsley and Cowtan, 2004; Emsley et al., 2010 https://www2.mrc-				

lmb.cam.ac.uk/personal/pemsley/coot/ Phenix Adams et al., 2010; Afonine et al., 2018; DiMaio et al., 2013 https://www.phenix-online.org/ Molprobity Chen et al., 2010 http://molprobity.biochem.duke.edu/ Image Studio LI-COR Biosciences https://www.licor.com/bio/image-studio/ Fiji/ImageJ Schindelin et al., 2012 https://imagej.net/Welcome GraphPad Prism version 8.0 GraphPad Software http://www.graphpad.com:443/ ImageQuant TL Toolbox version 8.2 Cytiva (formerly GE Healthcare) https://www.cytivalifesciences.com DeepEMhancer Sanchez-Garcia et al., 2020 http://www.biorxiv.org Other QUANTIFOIL(r) R1.2/1.3, 100 Holey Carbon Films, Grids: Cu 200 mesh Quantifoil Micro Tools GmbH https://www.quantifoil.com IMEM Thermo Fisher Cat# 12440-053 Resource availability Lead contact Information and requests for resources and reagents should be directed to the Lead Contact, Prof. Dr. Brenda Schulman (schulman@biochem.mpg.de).			
Protein levels of the substrates and a control protein DHFR were visualized by immunoblotting with anti-FLAG and anti-HA antibodies, respectively, and imaged using a Typhoon scanner (GE Healthcare).	anti-HA		
Ubiquitylation of the substrates was visualized by immunoblotting with anti-His antibody.	anti-His		
Fractions were collected from top of the gradient and separated by SDS-PAGE, followed by immunoblotting using the following antibodies: RMND5A (Santa Cruz), MAEA (R&D systems), RANBP9 (Novus Biologicals), TWA1 (Thermo Fisher), ARMC8 (Santa Cruz), WDR26 (Bethyl Laboratories), MKLN1 (Santa Cruz) and YPEL5 (Thermo Fisher).	WDR26		
	MKLN1		
Experimental Models: Cell Lines			
STARMethods Key resources table REAGENT or RESOURCE SOURCE IDENTIFIER Antibodies Monoclonal ANTI-FLAG M2 antibody Sigma Aldrich Cat# F1804; RRID: AB_262044 Anti-HA antibody produced in rabbit	K562		Suggestion: RRID:CVCL_0004(link)

Sigma Aldrich Cat# H6908; RRID: AB_260070 Goat anti-rabbit IgG Dylight488 conjugated Invitrogen Cat# 35552; RRID: AB_844398 Goat anti-mouse IgG Dylight633 conjugated Invitrogen Cat# 35512; RRID: AB_1307538 Anti-rabbit peroxidase antibody produced in goat Sigma Aldrich Cat# A9169; RRID: AB_258434 Anti-mouse IgG Peroxidase antibody produced in goat Sigma Aldrich Cat# A4416; RRID: AB_258167 Anti-His antibody produced in mouse Cell Signaling Technology Cat# 9991; RRID: AB_2797714 Goat polyclonal anti-RMND5A antibody Santa Cruz Cat# sc-161202, RRID: AB_2181510 Sheep polyclonal anti-MAEA antibody R&D Systems Cat# AF7288-SP, RRID: AB_10971438 Rabbit polyclonal anti-RANBP9 antibody Abnova Cat# PAB16671; RRID: AB_10677213 Rabbit polyclonal anti-TWA1 antibody Novus Cat# NBP1-32596; RRID: AB_2274921 Mouse monoclonal anti-ARMC8 antibody Santa Cruz Cat# sc-365307; RRID: AB_10850172 Mouse monoclonal anti-MKLN1 antibody Santa Cruz Cat# sc-398956; RRID: AB_2737249 Rabbit polyclonal anti-WDR26 antibody Bethyl Laboratories Cat# A302-245A; RRID: AB_1730876 Rabbit polyclonal anti-YPEL5 antibody Thermo Fisher Cat# PA5-26957; RRID: AB_2544457 Sheep polyclonal anti-hGid4 This study N/A HaloLink Resin Promega Cat# G1912 ANTI-FLAG M2 affinity gel Sigma Aldrich Cat# A2220 His-Select Nickel affinity gel Sigma Aldrich Cat# P6611 Glutathione Sepharose 4B GE Healthcare Cat# 17075605 StrepTactin Sepharose High Performance resin cytiva Cat# 28935599 His-Halo UBAUBQLN1 This study N/A Critical commercial assays EnzChek Phosphate Assay Kit ThermoFisher Scientific Cat# E6646 Bacterial and virus strains E. coli BL21 RIL (DE3) MPIB N/A E. coli DH5a MPIB N/A Chemicals, peptides, and recombinant proteins complete EDTA free Roche Cat# 05056489001 Aprotinin from				

bovine lung Sigma A1153-10MG Leupeptin Sigma L2884-250MG Benzamidine Sigma B6506-25G GGGGGFYVK-FAM MPIB N/A PTLVNGWPR MPIB N/A PTLVNGPRRDSTEGFTGRGWSGRGWSKGGK-FAM MPIB N/A PGLWRSPRRDSTEGFTGRGWSGRGWSKGGK-FAM MPIB N/A 3xFLAG peptide MPIB N/A Deposited data Apo Chelator-GIDSR4 This study EMDB: EMD-12541 Chelator-GIDSR4 + Fbp1 This study EMDB: EMD-12557 GIDSR4 This study EMDB: EMD-12548 SRS module This study EMDB: EMD-12559; PDB: 7NS3 Cat module This study EMDB: EMD-12560; PDB: 7NS4 SA module This study EMDB: EMD-12563; PDB: 7NSB Endogenous GIDAnt This study EMDB: EMD-12538 Endogenous Chelator-GIDAnt This study EMDB: EMD-12540 CTLHSR4 This study EMDB: EMD-12537 CTLHSR4 SRS module This study EMDB: EMD-12564; PDB: 7NSC CTLH-WDR26 SA and SRS modules This study EMDB: EMD-12545 CTLH-WDR26 supramolecular assembly This study EMDB: EMD-12542 CTLH-MKLN1 SA and SRS modules This study EMDB: EMD-12547 Fbp1 (crystal structure) This study PDB: 7NS5 Proteomics data This study PXD024462 Raw image data This study http://dx.doi.org/10.17632/rfpsg6939c.1 Experimental models: Cell lines Sf9 Insect cells Thermo Fisher Cat# 11496015 High Five Insect cells Thermo Fisher Cat# B85502 K562 human cells ATCC ATCC#CCL-243; RRID: CVCL_00004 Experimental models: Organisms/strains Saccharomyces cerevisiae: Strain S288C: BY4741; MATa his3D1leu2D0 met15D0 ura3D0 Euroscarf Cat# Y00000 CRLY12; BY4741, Gid4::KANMX This study N/A CRLY14; BY4741, Gid7::KANMX This study N/A CRLY45; BY4741, Gid8::Gid8-3xFLAG-KANMX Qiao et. al., 2020 N/A CRLY131; BY4741, Gid2::3xFLAG-Gid2 (K365A) Qiao et. al., 2020 N/A CRLY241; BY4741, Gid7::KANMX,				

Gid8::Gid8-3xFLAG-HPHNT1 This study N/A				
CRLY267; BY4741, Gid7::Gid7-3xHA-HPHNT1, Gid5::Gid5-3xFLAG-KANMX This study N/A				
CRLY498; BY4741, Fbp1::Fbp1-3xFLAG-HPHNT1, Pdr5::NATNT2 This study N/A				
CRLY504; BY4741, Fbp1::Fbp1-3xFLAG-HPHNT1, Pdr5::NATNT2, Gid7:KANMX This study N/A				
Recombinant DNA pCSJ95 Chen et al., 2017 N/A				
pCSJ125 Chen et al., 2017 N/A				
VBP6; pRS313-pGPD-Pck1-3xFLAG-CYC-pGPD-DHFR-HA-CYC This study N/A				
DSJC1; pRS313-pGPD-Fbp1-K32R/K35R-3xFLAG-CYC-pGPD-DHFR-HA-CYC This study N/A				
DSJC2; pRS313-pGPD-Fbp1-K280R/K281R-3xFLAG-CYC-pGPD-DHFR-HA-CYC This study N/A				
DSJC3; pRS313-pGPD-Fbp1-K32R/K35R/K280R/K281R-3xFLAG-CYC-pGPD-DHFR-HA-CYC This study N/A				
pRS415-pTEF-CYC This study N/A				
pRS415-pTEF-GFP-ScGid4-CYC This study N/A				
pLIB Gid1 This study N/A				
pLIB Gid2 This study N/A				
pLIB Gid4 This study N/A				
pLIB Gid5 This study N/A				
pLIB Gid7 This study N/A				
pLIB Gid8-TEV-2xStrep This study N/A				
pLIB Gid8 This study N/A				
pLIB Gid9 This study N/A				
pLIB RANBP9 This study N/A				
pLIB RMND5A This study N/A				
pLIB hGid4 This study N/A				
pLIB ARMC8 This study N/A				
pLIB 2xStrep-3C-ARMC8 This study N/A				
pLIB MAEA This study N/A				
pLIB WDR26 This study N/A				
pLIB GST-TEV-WDR26 This study N/A				
pLIB MKLN1 This study N/A				
pLIB YPEL5 This study N/A				
pLIB MAEA Y394A This study N/A				
pLIB RMND5A R340A This study N/A				
pLIB RMND5A I338A/L339A This study N/A				
pLIB GST-TEV-Uba1 This study N/A				
pBIG2 Gid1:Gid8-TEV-2xS:Gid5:Gid4:Gid2:Gid9 This study N/A				
pBIG2 Gid1:Gid8-TEV-2xS:Gid5:Gid2:Gid9 This study N/A				
pBIG2 Gid1:Gid8-TEV-2xS:Gid5:Gid2:Gid9:Gid7 This study N/A				
pBIG2 Gid1:Gid8-TEV-2xS:Gid5:Gid4:Gid2:Gid9:Gid7 This study N/A				
pBIG1				

RANBP9:TWA1-TEV-2xS:ARMC8 This study N/A pBIG1 RANBP9:TWA1:2xS-3C-ARMC8 This study N/A pBIG2 RANBP9:TWA1-TEV-2xS:ARMC8:RMND5A:MAEA This study N/A pBIG2 RANBP9:TWA1-TEV-2xS:ARMC8:RMND5A This study N/A pBIG2 RANBP9:TWA1-TEV-2xS:ARMC8:MAEA This study N/A pGEX GST-TEV-Gid4 (D1-116) This study N/A pGEX GST-TEV-Gid7 This study N/A pGEX GST-TEV-Gid7 (D1-285) This study N/A pGEX GST-TEV-hGid4 (D1-99) This study N/A pGEX GST-TEV-hGid4 (R189A) This study N/A pGEX GST-TEV-hGid4 (Y154A) This study N/A pGEX GST-TEV-hGid4 (F174A) This study N/A pGEX GST-TEV-hGid4 (F239A) This study N/A pGEX GST-TEV-hGid4 (C156D) This study N/A pGEX GST-TEV-hGid4 (E298A) This study N/A pGEX GST-TEV-hGid4 (H147D) This study N/A pGEX GST-TEV-hGid4 (Y158A/F174A) This study N/A pGEX GST-TEV-hGid4 (Y158A/F239A) This study N/A pGEX GST-TEV-hGid4 (F174A/F239A) This study N/A pGEX GST-TEV-hGid4 (Y297A/F229A) This study N/A pGEX GST-TEV-hGid4 (D297-300) This study N/A pRSF Ubc8-6xHis This study N/A pRSF Fbp1-6xHis This study N/A pRSF Fbp1 (K32R/K35R)-6xHis This study N/A pRSF Fbp1 (K280R/K281R)-6xHis This study N/A pRSF Fbp1 (K32R/K35R/K280R/K281R)-6xHis This study N/A pRSF Fbp1-GGGGS-sortag-6xHis This study N/A pRSF Mdh2-GGGGS-sortag-6xHis This study N/A pRSF Pck1-GGGGS-sortag-6xHis This study N/A pQlink Fbp1-TEV-V5-2xS This study N/A pRSF Ube2H-6xHis This study N/A pGEX GST-3C-Ub This study N/A pGEX GST-3C-Ub K0 (all K > R) This study N/A pGEX GST-3C-Ub K6 (all K > R; R6K) This study N/A pGEX GST-3C-Ub K11 (all K > R; R11K) This study N/A pGEX GST-3C-Ub K27 (all K > R; R27K) This study N/A pGEX GST-3C-Ub K29 (all K > R; R29K) This study				

N/A pGEX GST-3C-Ub K33 (all K > R; R33K) This study N/A pGEX GST-3C-Ub K48 (all K > R; R48K) This study N/A pGEX GST-3C-Ub K63 (all K > R; R63K) This study N/A pET3b Ub This study N/A pET29 sortase A Chen et al., 2011b N/A Software and algorithms FOCUS Biyani et al., 2017 https://focus.c-cina.unibas.ch/documentation.php SerialEM Mastronarde, 2003 https://bio3d.colorado.edu/SerialEM/ MOTIONCOR2 Zheng et al., 2017 https://emcore.ucsf.edu/ucsf-software Gctf Zhang, 2016 https://www2.mrc-lmb.cam.ac.uk/download/gctf/ Gautomatch Kai Zhang https://www2.mrc-lmb.cam.ac.uk/download/gautomatch-053/ Relion3.0/3.1 Fernandez-Leiro and Scheres, 2017; Scheres, 2012; Zivanov et al., 2018 https://www3.mrc-lmb.cam.ac.uk/relion/index.php/Main_Page Phyre2 Kelley et al., 2015 http://www.sbg.bio.ic.ac.uk/~phyre2/html/page.cgi?id=index SWISS-MODEL Waterhouse et al., 2018 https://swissmodel.expasy.org UCSF Chimera Pettersen et al., 2004 https://www.cgl.ucsf.edu/chimera/ UCSF ChimeraX Goddard et al., 2018 https://www.rbvi.ucsf.edu/chimerax/ PyMOL v2.1 Schrodinger https://pymol.org/2/ CCP-EM Burnley et al., 2017 https://www.ccpem.ac.uk/download.php Buccaneer Cowtan, 2006 http://www.ysbl.york.ac.uk/~cowtan/buccaneer/buccaneer.html Coot Emsley and Cowtan, 2004; Emsley et al., 2010 https://www2.mrc-lmb.cam.ac.uk/personal/pemsley/coot/ Phenix Adams et al., 2010; Afonine et al., 2018; DiMaio et al., 2013 https://www.phenix-online.org/ Molprobity Chen et al., 2010 http://molprobity.biochem.duke.edu/ Image Studio LI-COR Biosciences https://www.licor.com/bio/image-studio/ Fiji/ImageJ Schindelin et al., 2012 https://imagej.net/Welcome GraphPad Prism version 8.0 GraphPad Software http://www.graphpad.com:443/ ImageQuant TL Toolbox version 8.2 Cytiva (formerly GE Healthcare) https://www.cytivalifesciences.com DeepEMhancer Sanchez-Garcia et				

al., 2020 http://www.biorxiv.org Other QUANTIFOIL(r) R1.2/1.3, 100 Holey Carbon Films, Grids: Cu 200 mesh Quantifoil Micro Tools GmbH https://www.quantifoil.com IMEM Thermo Fisher Cat# 12440-053 Resource availability Lead contact Information and requests for resources and reagents should be directed to the Lead Contact, Prof. Dr. Brenda Schulman (schulman@biochem.mpg.de).			
Experimental Models: Organisms/Strains			
Data and code availability The accession codes for the PDB models and EM maps are available in RCSB and EMDB, respectively, as follows: Apo Chelator-GIDSR4, EMDB: EMD-12541; Chelator-GIDSR4 + Fbp1, EMDB: EMD-12557; GIDSR4, EMDB: EMD-12548; SRS module, EMDB: EMD-12559, PDB: 7NS3; Cat module, EMDB: EMD-12560; PDB: 7NS4; SA module, EMDB: EMD-12563; PDB: 7NSB; Endogenous GIDAnt, EMDB: EMD-12538; Endogenous Chelator-GIDAnt, EMDB: EMD-12540; CTLHSR4, EMDB: EMD-12537; CTLHSR4 SRS module, EMDB:EMD-12564; PDB: 7NSC; CTLH-WDR26 supramolecular assembly, EMDB: EMD-12542; CTLH-WDR26 SA and SRS modules, EMDB: EMD-12545; CTLH-MKLN1 SA and SRS modules, EMDB: EMD-12547; Fbp1 crystal structure, PDB: 7NS5.	Chelator-GIDSR4 + Fbp1		
Recombinant DNA			
To test if overexpression of Gid4 affects degradation of Fbp1 in DGid7 yeast (Figure S1E), the GFP-Gid4 overexpression plasmid was transformed together with the Fbp1-3xFLAG PRT plasmid into different yeast strains (WT, DGid7 and a Gid2K365A catalytically inactive mutant).	Fbp1-3xFLAG PRT		
Sense and antisense gRNA were cloned into pBABED-U6-Puromycin plasmid (gift from Thomas Macartney, University of Dundee, UK) and pX335-Cas9(D10A) (Addgene) (Cong et al., 2013), respectively.	pBABED-U6-Puromycin		

Software and Algorithms			
STARMethods Key resources table REAGENT or RESOURCE SOURCE IDENTIFIER Antibodies Monoclonal ANTI-FLAG M2 antibody Sigma Aldrich Cat# F1804; RRID: AB_262044 Anti-HA antibody produced in rabbit Sigma Aldrich Cat# H6908; RRID: AB_260070 Goat anti-rabbit IgG Dylight488 conjugated Invitrogen Cat# 35552; RRID: AB_844398 Goat anti-mouse IgG Dylight633 conjugated Invitrogen Cat# 35512; RRID: AB_1307538 Anti-rabbit peroxidase antibody produced in goat Sigma Aldrich Cat# A9169; RRID: AB_258434 Anti-mouse IgG Peroxidase antibody produced in goat Sigma Aldrich Cat# A4416; RRID: AB_258167 Anti-His antibody produced in mouse Cell Signaling Technology Cat# 9991; RRID: AB_2797714 Goat polyclonal anti-RMND5A antibody Santa Cruz Cat# sc-161202, RRID: AB_2181510 Sheep polyclonal anti-MAEA antibody R&D Systems Cat# AF7288-SP, RRID: AB_10971438 Rabbit polyclonal anti-RANBP9 antibody Abnova Cat# PAB16671; RRID: AB_10677213 Rabbit polyclonal anti-TWA1 antibody Novus Cat# NBP1-32596; RRID: AB_2274921 Mouse monoclonal anti-ARMC8 antibody Santa Cruz Cat# sc-365307; RRID: AB_10850172 Mouse monoclonal anti-MKLN1 antibody Santa Cruz Cat# sc-398956; RRID: AB_2737249 Rabbit polyclonal anti-WDR26 antibody Bethyl Laboratories Cat# A302-245A; RRID: AB_1730876 Rabbit polyclonal anti-YPEL5 antibody Thermo Fisher Cat# PA5-26957; RRID: AB_2544457 Sheep polyclonal anti-hGid4 This study N/A HaloLink Resin Promega Cat# G1912 ANTI-FLAG M2 affinity gel Sigma Aldrich Cat# A2220 His-Select Nickel affinity gel Sigma Aldrich Cat# P6611 Glutathione Sepharose 4B GE Healthcare Cat# 17075605 StrepTactin Sepharose High Performance resin cytiva Cat# 28935599 His-Halo UBAUBQLN1	Thermo Fisher Cat#		
	PyMOL		Suggestion: PyMOL, RRID:SCR_000305(link)
	https://www2.mrc-lmb.cam.ac.uk/personal/pemsley/coot/		
	https://www.phenix-online.org/		
	Molprobity		
	http://molprobity.biochem.duke.edu/		
	GraphPad		Suggestion: GraphPad Prism, RRID:SCR_002798(link)
	http://www.graphpad.com:443/		
	ImageQuant		

This study N/A Critical commercial assays EnzChek Phosphate Assay Kit ThermoFisher Scientific Cat# E6646 Bacterial and virus strains E. coli BL21 RIL (DE3) MPIB N/A E. coli DH5a MPIB N/A Chemicals, peptides, and recombinant proteins complete EDTA free Roche Cat# 05056489001 Aprotinin from bovine lung Sigma A1153-10MG Leupeptin Sigma L2884-250MG Benzamidine Sigma B6506-25G GGGGGFYVK-FAM MPIB N/A PTLVNGWPR MPIB N/A PTLVNGPRRDSTEGFTGRGWSGRGWSKGGK-FAM MPIB N/A PGLWRSPRRDSTEGFTGRGWSGRGWSKGGK-FAM MPIB N/A 3xFLAG peptide MPIB N/A Deposited data Apo Chelator-GIDSR4 This study EMDB: EMD-12541 Chelator-GIDSR4 + Fbp1 This study EMDB: EMD-12557 GIDSR4 This study EMDB: EMD-12548 SRS module This study EMDB: EMD-12559; PDB: 7NS3 Cat module This study EMDB: EMD-12560; PDB: 7NS4 SA module This study EMDB: EMD-12563; PDB: 7NSB Endogenous GIDAnt This study EMDB: EMD-12538 Endogenous Chelator-GIDAnt This study EMDB: EMD-12540 CTLHSR4 This study EMDB: EMD-12537 CTLHSR4 SRS module This study EMDB: EMD-12564; PDB: 7NSC CTLH-WDR26 SA and SRS modules This study EMDB: EMD-12545 CTLH-WDR26 supramolecular assembly This study EMDB: EMD-12542 CTLH-MKLN1 SA and SRS modules This study EMDB: EMD-12547 Fbp1 (crystal structure) This study PDB: 7NS5 Proteomics data This study PXD024462 Raw image data This study http://dx.doi.org/10.17632/rfpsg6939c.1 Experimental models: Cell lines Sf9 Insect cells Thermo Fisher Cat# 11496015 High Five Insect cells Thermo Fisher Cat# B85502 K562 human cells ATCC ATCC#CCL-243; RRID: CVCL_00004 Experimental models: Organisms/strains Saccharomyces cerevisiae: Strain S288C: BY4741; MATa his3D1leu2D0 met15D0 ura3D0 Euroscarf Cat# Y00000				

CRLY12; BY4741, Gid4::KANMX This study N/A CRLY14; BY4741, Gid7::KANMX This study N/A CRLY45; BY4741, Gid8::Gid8-3xFLAG-KANMX Qiao et. al., 2020 N/A CRLY131; BY4741, Gid2::3xFLAG-Gid2 (K365A) Qiao et. al., 2020 N/A CRLY241; BY4741, Gid7::KANMX, Gid8::Gid8-3xFLAG-HPHNT1 This study N/A CRLY267; BY4741, Gid7::Gid7-3xHA-HPHNT1, Gid5::Gid5-3xFLAG-KANMX This study N/A CRLY498; BY4741, Fbp1::Fbp1-3xFLAG-HPHNT1, Pdr5::NATNT2 This study N/A CRLY504; BY4741, Fbp1::Fbp1-3xFLAG-HPHNT1, Pdr5::NATNT2, Gid7:KANMX This study N/A Recombinant DNA pCSJ95 Chen et al., 2017 N/A pCSJ125 Chen et al., 2017 N/A VBP6; pRS313-pGPD-Pck1-3xFLAG-CYC-pGPD-DHFR-HA-CYC This study N/A DSJC1; pRS313-pGPD-Fbp1-K32R/K35R-3xFLAG-CYC-pGPD-DHFR-HA-CYC This study N/A DSJC2; pRS313-pGPD-Fbp1-K280R/K281R-3xFLAG-CYC-pGPD-DHFR-HA-CYC This study N/A DSJC3; pRS313-pGPD-Fbp1-K32R/K35R/K280R/K281R-3xFLAG-CYC-pGPD-DHFR-HA-CYC This study N/A pRS415-pTEF-CYC This study N/A pRS415-pTEF-GFP-ScGid4-CYC This study N/A pLIB Gid1 This study N/A pLIB Gid2 This study N/A pLIB Gid4 This study N/A pLIB Gid5 This study N/A pLIB Gid7 This study N/A pLIB Gid8-TEV-2xStrep This study N/A pLIB Gid8 This study N/A pLIB Gid9 This study N/A pLIB RANBP9 This study N/A pLIB RMND5A This study N/A pLIB hGid4 This study N/A pLIB ARMC8 This study N/A pLIB 2xStrep-3C-ARMC8 This study N/A pLIB MAEA This study N/A pLIB WDR26 This study N/A pLIB GST-TEV-WDR26 This study N/A pLIB MKLN1 This study N/A pLIB YPEL5 This study N/A pLIB MAEA Y394A This study N/A pLIB RMND5A R340A This study N/A pLIB RMND5A I338A/L339A This study N/A pLIB GST-TEV-Uba1				

This study N/A pBIG2 Gid1:Gid8-TEV-2xS:Gid5:Gid4:Gid2:Gid9 This study N/A pBIG2 Gid1:Gid8-TEV-2xS:Gid5:Gid2:Gid9 This study N/A pBIG2 Gid1:Gid8-TEV-2xS:Gid5:Gid2:Gid9:Gid7 This study N/A pBIG2 Gid1:Gid8-TEV-2xS:Gid5:Gid4:Gid2:Gid9:Gid7 This study N/A pBIG1 RANBP9:TWA1-TEV-2xS:ARMC8 This study N/A pBIG1 RANBP9:TWA1:2xS-3C-ARMC8 This study N/A pBIG2 RANBP9:TWA1-TEV-2xS:ARMC8:RMND5A:MAEA This study N/A pBIG2 RANBP9:TWA1-TEV-2xS:ARMC8:RMND5A This study N/A pBIG2 RANBP9:TWA1-TEV-2xS:ARMC8:MAEA This study N/A pGEX GST-TEV-Gid4 (D1-116) This study N/A pGEX GST-TEV-Gid7 This study N/A pGEX GST-TEV-Gid7 (D1-285) This study N/A pGEX GST-TEV-hGid4 (D1-99) This study N/A pGEX GST-TEV-hGid4 (R189A) This study N/A pGEX GST-TEV-hGid4 (Y154A) This study N/A pGEX GST-TEV-hGid4 (F174A) This study N/A pGEX GST-TEV-hGid4 (F239A) This study N/A pGEX GST-TEV-hGid4 (C156D) This study N/A pGEX GST-TEV-hGid4 (E298A) This study N/A pGEX GST-TEV-hGid4 (H147D) This study N/A pGEX GST-TEV-hGid4 (Y158A/F174A) This study N/A pGEX GST-TEV-hGid4 (Y158A/F239A) This study N/A pGEX GST-TEV-hGid4 (F174A/F239A) This study N/A pGEX GST-TEV-hGid4 (Y297A/F229A) This study N/A pGEX GST-TEV-hGid4 (D297-300) This study N/A pRSF Ubc8-6xHis This study N/A pRSF Fbp1-6xHis This study N/A pRSF Fbp1 (K32R/K35R)-6xHis This study N/A pRSF Fbp1 (K280R/K281R)-6xHis This study N/A pRSF Fbp1 (K32R/K35R/K280R/K281R)-6xHis This study N/A pRSF Fbp1-GGGGS-sortag-6xHis This study N/A pRSF Mdh2-GGGGS-sortag-6xHis This study N/A pRSF Pck1-GGGGS-sortag-6xHis This study N/A pQlink Fbp1-TEV-V5-2xS This study N/A pRSF Ube2H-6xHis This study N/				

A pGEX GST-3C-Ub This study N/A pGEX GST-3C-Ub K0 (all K > R) This study N/A pGEX GST-3C-Ub K6 (all K > R; R6K) This study N/A pGEX GST-3C-Ub K11 (all K > R; R11K) This study N/A pGEX GST-3C-Ub K27 (all K > R; R27K) This study N/A pGEX GST-3C-Ub K29 (all K > R; R29K) This study N/A pGEX GST-3C-Ub K33 (all K > R; R33K) This study N/A pGEX GST-3C-Ub K48 (all K > R; R48K) This study N/A pGEX GST-3C-Ub K63 (all K > R; R63K) This study N/A pET3b Ub This study N/A pET29 sortase A Chen et al., 2011b N/A Software and algorithms FOCUS Biyani et al., 2017 https://focus.c-cina.unibas.ch/documentation.php SerialEM Mastronarde, 2003 https://bio3d.colorado.edu/SerialEM/ MOTIONCOR2 Zheng et al., 2017 https://emcore.ucsf.edu/ucsf-software Gctf Zhang, 2016 https://www2.mrc-lmb.cam.ac.uk/download/gctf/ Gautomatch Kai Zhang https://www2.mrc-lmb.cam.ac.uk/download/gautomatch-053/ Relion3.0/3.1 Fernandez-Leiro and Scheres, 2017; Scheres, 2012; Zivanov et al., 2018 https://www3.mrc-lmb.cam.ac.uk/relion/index.php/Main_Page Phyre2 Kelley et al., 2015 http://www.sbg.bio.ic.ac.uk/~phyre2/html/page.cgi?id=index SWISS-MODEL Waterhouse et al., 2018 https://swissmodel.expasy.org UCSF Chimera Pettersen et al., 2004 https://www.cgl.ucsf.edu/chimera/ UCSF ChimeraX Goddard et al., 2018 https://www.rbvi.ucsf.edu/chimerax/ PyMOL v2.1 Schrodinger https://pymol.org/2/ CCP-EM Burnley et al., 2017 https://www.ccpem.ac.uk/download.php Buccaneer Cowtan, 2006 http://www.ysbl.york.ac.uk/~cowtan/buccaneer/buccaneer.html Coot Emsley and Cowtan, 2004; Emsley et al., 2010 https://www2.mrc-lmb.cam.ac.uk/personal/pemsley/coot/ Phenix Adams et al., 2010; Afonine et al., 2018; DiMaio et al., 2013 https://www.phenix-online.org/ Molprobity Chen et al., 2010 http://molprobity.biochem.duke.edu/ Image Studio LI-COR Biosciences https://				

www.licor.com/bio/image-studio/ Fiji/ImageJ Schindelin et al., 2012 https://imagej.net/Welcome GraphPad Prism version 8.0 GraphPad Software http://www.graphpad.com:443/ ImageQuant TL Toolbox version 8.2 Cytiva (formerly GE Healthcare) https://www.cytivalifesciences.com DeepEMhancer Sanchez-Garcia et al., 2020 http://www.biorxiv.org Other QUANTIFOIL(r) R1.2/1.3, 100 Holey Carbon Films, Grids: Cu 200 mesh Quantifoil Micro Tools GmbH https://www.quantifoil.com IMEM Thermo Fisher Cat# 12440-053 Resource availability Lead contact Information and requests for resources and reagents should be directed to the Lead Contact, Prof. Dr. Brenda Schulman (schulman@biochem.mpg.de).			
All proteomics data have been deposited on ProteomeXchange with the dataset identifier PRIDE: PXD024462.	ProteomeXchange		
The values of A360 obtained for the buffer-only control were subtracted from all the experimental measurements, which were then plotted in GraphPad Prism.	GraphPad Prism		
The Perseus software package version 1.6.0.7 was used for the data analysis (Tyanova et al., 2016).	Perseus		
Raw MS data were searched against UniProt Yeast FASTA using MaxQuant (version 1.6.2.10) with a 1% FDR at peptide and protein level.	MaxQuant		
Cell culture and generation of CRISPR-Cas9 knock out cell lines K562 erythroleukemia cell line was obtained from ATCC (CCL-243(tm)) and cultured in IMDM completed with 10% (v/v) FBS (GIBCO) and antibiotics (100 units/ml penicillin, 0.1 mg/ml streptomycin, GIBCO).	GIBCO		
Fractions were collected from top of the gradient and separated by SDS-PAGE, followed by immunoblotting using the following antibodies: RMND5A (Santa Cruz), MAEA (R&D systems), RANBP9 (Novus Biologicals), TWA1 (Thermo Fisher), ARMC8 (Santa Cruz), WDR26 (Bethyl Laboratories), MKLN1 (Santa Cruz) and YPEL5 (Thermo Fisher).	Novus Biologicals		

Most of ARMC8 was built automatically with Buccaneer (Cowtan, 2006) and refined manually with Coot.	Coot		
All the models were subjected to iterative rounds of manual building with Coot and real space refinement in PHENIX (Adams et al., 2010; Afonine et al., 2018; DiMaio et al., 2013) until a satisfactory model quality, in terms of its geometry and agreement with the map, was obtained.	PHENIX		
Quantification and statistical analysis For the assays described in the section "In vivo yeast substrate degradation assays," protein bands visualized by western blots were quantified using ImageStudioLite software (Li-Cor).	ImageStudioLite		

Other Entities Detected

Your Sentences	Recognized Entity
Unresolved RRIDs	
STARMethods Key resources table REAGENT or RESOURCE SOURCE IDENTIFIER Antibodies Monoclonal ANTI-FLAG M2 antibody Sigma Aldrich Cat# F1804; RRID: AB_262044 Anti-HA antibody produced in rabbit Sigma Aldrich Cat# H6908; RRID: AB_260070 Goat anti-rabbit IgG Dylight488 conjugated Invitrogen Cat# 35552; RRID: AB_844398 Goat anti-mouse IgG Dylight633 conjugated Invitrogen Cat# 35512; RRID: AB_1307538 Anti-rabbit peroxidase antibody produced in goat Sigma Aldrich Cat# A9169; RRID: AB_258434 Anti-mouse IgG Peroxidase antibody produced in goat Sigma Aldrich Cat# A4416; RRID: AB_258167 Anti-His antibody produced in mouse Cell Signaling Technology Cat# 9991; RRID: AB_2797714 Goat polyclonal anti-RMND5A antibody Santa Cruz Cat# sc-161202, RRID: AB_2181510 Sheep polyclonal anti-MAEA antibody R&D Systems Cat# AF7288-SP, RRID: AB_10971438 Rabbit polyclonal anti-RANBP9 antibody Abnova Cat# PAB16671; RRID: AB_10677213 Rabbit polyclonal anti-TWA1 antibody Novus Cat# NBP1-32596; RRID: AB_2274921 Mouse monoclonal anti-ARMC8 antibody Santa Cruz Cat# sc-365307; RRID: AB_10850172 Mouse monoclonal anti-MKLN1 antibody Santa Cruz Cat# sc-398956; RRID: AB_2737249 Rabbit polyclonal anti-WDR26 antibody Bethyl Laboratories Cat# A302-245A; RRID: AB_1730876 Rabbit polyclonal anti-YPEL5 antibody Thermo Fisher Cat# PA5-26957; RRID: AB_2544457 Sheep polyclonal anti-hGid4 This study N/A HaloLink Resin Promega Cat# G1912 ANTI-FLAG M2 affinity gel Sigma Aldrich Cat# A2220 His-Select Nickel affinity gel Sigma Aldrich Cat# P6611 Glutathione Sepharose 4B GE Healthcare Cat# 17075605 StrepTactin Sepharose High Performance resin cytiva Cat# 28935599 His-Halo UBAUBQLN1 This study N/A Critical commercial assays EnzChek Phosphate Assay Kit ThermoFisher Scientific Cat# E6646 Bacterial and virus strains E. coli BL21 RIL (DE3) MPIB N/A E. coli DH5a MPIB N/A Chemicals, peptides, and recombinant proteins complete EDTA free Roche Cat#	RRID: CVCL_00004

05056489001 Aprotinin from bovine lung Sigma A1153-10MG Leupeptin Sigma L2884-250MG Benzamidine Sigma B6506-25G GGGGGFYVK-FAM MPIB N/A PTLVNGWPR MPIB N/A PTLVNGPRRDSTEGFTGRGWSGRGWSKGGK-FAM MPIB N/A PGLWRSPRRDSTEGFTGRGWSGRGWSKGGK-FAM MPIB N/A 3xFLAG peptide MPIB N/A Deposited data Apo Chelator-GIDSR4 This study EMDB: EMD-12541 Chelator-GIDSR4 + Fbp1 This study EMDB: EMD-12557 GIDSR4 This study EMDB: EMD-12548 SRS module This study EMDB: EMD-12559; PDB: 7NS3 Cat module This study EMDB: EMD-12560; PDB: 7NS4 SA module This study EMDB: EMD-12563; PDB: 7NSB Endogenous GIDAnt This study EMDB: EMD-12538 Endogenous Chelator-GIDAnt This study EMDB: EMD-12540 CTLHSR4 This study EMDB: EMD-12537 CTLHSR4 SRS module This study EMDB: EMD-12564; PDB: 7NSC CTLH-WDR26 SA and SRS modules This study EMDB: EMD-12545 CTLH-WDR26 supramolecular assembly This study EMDB: EMD-12542 CTLH-MKLN1 SA and SRS modules This study EMDB: EMD-12547 Fbp1 (crystal structure) This study PDB: 7NS5 Proteomics data This study PXD024462 Raw image data This study http://dx.doi.org/10.17632/rfpsg6939c.1 Experimental models: Cell lines Sf9 Insect cells Thermo Fisher Cat# 11496015 High Five Insect cells Thermo Fisher Cat# B85502 K562 human cells ATCC ATCC#CCL-243; RRID: CVCL_00004 Experimental models: Organisms/strains Saccharomyces cerevisiae: Strain S288C: BY4741; MATa his3D1leu2D0 met15D0 ura3D0 Euroscarf Cat# Y00000 CRLY12; BY4741, Gid4::KANMX This study N/A CRLY14; BY4741, Gid7::KANMX This study N/A CRLY45; BY4741, Gid8::Gid8-3xFLAG-KANMX Qiao et. al., 2020 N/A CRLY131; BY4741, Gid2::3xFLAG-Gid2 (K365A) Qiao et. al., 2020 N/A CRLY241; BY4741, Gid7::KANMX, Gid8::Gid8-3xFLAG-HPHNT1 This study N/A CRLY267; BY4741, Gid7::Gid7-3xHA-HPHNT1, Gid5::Gid5-3xFLAG-KANMX This study N/A CRLY498; BY4741, Fbp1::Fbp1-3xFLAG-HPHNT1, Pdr5::NATNT2 This study N/A CRLY504; BY4741, Fbp1::Fbp1-3xFLAG-HPHNT1,		

Pdr5::NATNT2, Gid7:KANMX This study N/A Recombinant DNA pCSJ95 Chen et al., 2017 N/A pCSJ125 Chen et al., 2017 N/A VBP6; pRS313-pGPD-Pck1-3xFLAG-CYC-pGPD-DHFR-HA-CYC This study N/A DSJC1; pRS313-pGPD-Fbp1-K32R/K35R-3xFLAG-CYC-pGPD-DHFR-HA-CYC This study N/A DSJC2; pRS313-pGPD-Fbp1-K280R/K281R-3xFLAG-CYC-pGPD-DHFR-HA-CYC This study N/A DSJC3; pRS313-pGPD-Fbp1-K32R/K35R/K280R/K281R-3xFLAG-CYC-pGPD-DHFR-HA-CYC This study N/A pRS415-pTEF-CYC This study N/A pRS415-pTEF-GFP-ScGid4-CYC This study N/A pLIB Gid1 This study N/A pLIB Gid2 This study N/A pLIB Gid4 This study N/A pLIB Gid5 This study N/A pLIB Gid7 This study N/A pLIB Gid8-TEV-2xStrep This study N/A pLIB Gid8 This study N/A pLIB Gid9 This study N/A pLIB RANBP9 This study N/A pLIB RMND5A This study N/A pLIB hGid4 This study N/A pLIB ARMC8 This study N/A pLIB 2xStrep-3C-ARMC8 This study N/A pLIB MAEA This study N/A pLIB WDR26 This study N/A pLIB GST-TEV-WDR26 This study N/A pLIB MKLN1 This study N/A pLIB YPEL5 This study N/A pLIB MAEA Y394A This study N/A pLIB RMND5A R340A This study N/A pLIB RMND5A I338A/L339A This study N/A pLIB GST-TEV-Uba1 This study N/A pBIG2 Gid1:Gid8-TEV-2xS:Gid5:Gid4:Gid2:Gid9 This study N/A pBIG2 Gid1:Gid8-TEV-2xS:Gid5:Gid2:Gid9 This study N/A pBIG2 Gid1:Gid8-TEV-2xS:Gid5:Gid2:Gid9:Gid7 This study N/A pBIG2 Gid1:Gid8-TEV-2xS:Gid5:Gid4:Gid2:Gid9:Gid7 This study N/A pBIG1 RANBP9:TWA1-TEV-2xS:ARMC8 This study N/A pBIG1 RANBP9:TWA1:2xS-3C-ARMC8 This study N/A pBIG2 RANBP9:TWA1-TEV-2xS:ARMC8:RMND5A:MAEA This study N/A pBIG2 RANBP9:TWA1-TEV-2xS:ARMC8:RMND5A This study N/A pBIG2 RANBP9:TWA1-TEV-2xS:ARMC8:MAEA This study N/A pGEX GST-TEV-Gid4 (D1-116) This study N/A pGEX GST-TEV-Gid7 This study N/A pGEX GST-TEV-Gid7 (D1-285) This study N/A pGEX GST-TEV-hGid4 (D1-99) This study N/A pGEX GST-TEV-hGid4 (R189A) This study N/A pGEX GST-TEV-hGid4 (Y154A) This study N/A pGEX GST-TEV-hGid4 (F174A) This		

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